



PAKISTAN'S FIRST AI COMMERCE OS

Platform & Courier API Integration R&D Report

A comprehensive technical research & analysis covering Daraz, Shopify, TCS Courier, Leopards Courier, M&P Courier, Pakistan Post, and DHL Express — including API authentication, endpoints, capabilities, limitations, and integration recommendations.

Daraz

Shopify

TCS Courier

Leopards

M&P

Pakistan Post

DHL Express

REPORT DATE

July 2026

VERSION

1.0.0

PLATFORMS COVERED

7 Integrations

PREPARED FOR

Zyro Engineering

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This report provides a comprehensive technical analysis of seven commerce API integrations relevant to the Zyro Commerce OS platform. It covers two e-commerce marketplace platforms (Daraz and Shopify) and five courier/logistics providers (TCS, Leopards, M&P, Pakistan Post, and DHL Express).

Each integration is analysed across authentication method, available API endpoints, data schema, rate limits, webhook support, known limitations, and implementation complexity. The report concludes with a unified comparison matrix and actionable integration recommendations for the Zyro engineering team.

7

Integrations
Analysed

2

Marketplace
Platforms

5

Courier Providers

3

Auth Methods
Used

2

Webhook-Capable
APIs

WEBHOOK-ENABLED PLATFORMS

Shopify, WooCommerce (Daraz has no webhook support)

AUTHENTICATION METHODS FOUND

HMAC-SHA256 signing, Bearer Token, API Key header, Body credentials, OAuth 2.0

PAKISTAN-NATIVE APIS

TCS, Leopards, M&P, Pakistan Post, Daraz Pakistan

INTERNATIONAL APIS

Shopify (Global), DHL Express (Global)

⚠️ KEY FINDING — CUSTOMER PRIVACY (DARAZ)

Daraz's DataMoat system masks customer names and phone numbers for all sellers by default. To access real customer data, sellers must apply for the **Sensitive Data Privilege** in the Daraz App Console — approval is not guaranteed and may take 2–4 weeks. This is a critical limitation for CRM features.

✓ RECOMMENDED PRIORITY ORDER

For Zyro MVP launch: **Shopify** → **TCS** → **Leopards** → **Daraz** → **DHL** → **M&P** → **Pakistan Post**. Shopify has the cleanest API with full webhook support. TCS and Leopards are the dominant domestic couriers. Daraz can follow after the DataMoat privilege is secured. Pakistan Post is tracking-only with no merchant booking API.



Daraz Open Platform

E-Commerce Marketplace · Pakistan, Bangladesh, Sri Lanka, Nepal, Myanmar

NO WEBHOOKS

PULL ONLY

HMAC-SHA256

DEVELOPER PORTAL

open.daraz.com

PAKISTAN BASE URL

`https://api.daraz.pk/rest`

AUTH METHOD

HMAC-SHA256 Request
Signing

PLATFORM ID (ZYRO DB)

5

MAX ORDERS PER REQUEST

100 (offset-based pagination)

WEBHOOK SUPPORT

None — pull only

REQUIRED CREDENTIALS

- `daraz_app_key` → App Key from open.daraz.com App Console (numeric, e.g. 123456)
- `daraz_app_secret` → App Secret from App Console (32-char hex string)
- `daraz_access_token` → Access Token from Self-Authorization (long alphanumeric, expires or permanent depending on app type)

Signing Algorithm

Collect all
params

Sort keys A→Z

Concat apiPath +
key1val1key2val2...

HMAC-SHA256
with appSecret

Uppercase HEX

KEY API ENDPOINTS

GET

`/orders/get`

List orders with filters: status, date range, offset, limit (max 100)

GET

`/orders/items/get`

Get line items for a specific order (order items not included in `/orders/get`)

GET

`/products/get`

List products with categories, SKUs, inventory levels

GET

`/order/get`

Get a single order by order_id

POST

`/orders/packing/set`

Mark an order as packed (triggers fulfillment workflow in Daraz)

POST

`/orders/rts/set`

Mark items as Ready-To-Ship (Daraz logistics only)

GET

`/finance/transaction/details/get`

Settlement transactions and payment records

🕒 RATE LIMITS

No officially published rate limit, but Daraz throttles aggressively at approximately **60 requests/minute**. For bulk syncs use offset-based pagination (step 100) and add 1-second delays between batches. Daraz returns HTTP 429 with a message in the JSON body (not standard headers).

⚠ DATAMOAT PRIVACY SYSTEM

Customer **names and phone numbers are masked** by default. The API returns values like "****" or empty strings. Apply for **Sensitive Data Privilege (SDP)** in the App Console. Approval requires business verification and may take 2–4 weeks. Without SDP, CRM features that rely on customer PII cannot function.

STRENGTHS

- ✓ Dominant marketplace in Pakistan with high order volume
- ✓ Covers 5 countries in South/Southeast Asia
- ✓ Rich product catalogue and inventory APIs
- ✓ Financial settlement data accessible via API
- ✓ HMAC signing is well-documented

LIMITATIONS

- ✗ No webhook support — pull-only, requires polling
- ✗ Customer PII masked by DataMoat (SDP required)
- ✗ Order items require a separate API call
- ✗ Search limited to Order ID only (no name/phone search)
- ✗ Access Token can expire (varies per app type)





Shopify Admin API

E-Commerce Platform · Global (195+ countries)

WEBHOOKS ✓

PUSH + PULL

TOKEN AUTH

API VERSION

2024-01 (REST) /
2024-01 (GraphQL)

BASE URL PATTERN

https://{store}.myshopify.com/admin/api/2024-01

AUTH METHOD

X-Shopify-Access-Token header

PLATFORM ID (ZYRO DB)

6

MAX ORDERS PER REQUEST

250 (cursor-based pagination)

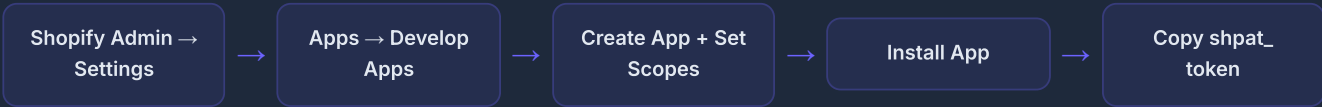
WEBHOOK SUPPORT

Yes — HMAC-SHA256 verified

REQUIRED CREDENTIALS

- shopify_store_domain → mystore.myshopify.com (without https://)
- shopify_access_token → Admin API access token (starts with shpat_) — shown once after app install
- shopify_webhook_secret → Optional but recommended — used to verify HMAC-SHA256 webhook signatures

Shopify App Setup Flow (Custom App)



KEY API ENDPOINTS

GET	/orders.json	List orders (max 250). Params: status, created_at_min/max, limit, since_id, fields
GET	/orders/count.json	Get total order count — much faster than fetching orders for stats
GET	/orders/{id}.json	Get single order with all line items, customer, shipping, billing included
GET	/customers.json	List customers with full PII (name, phone, email, address)
GET	/products.json	List products with variants, SKUs, inventory levels
POST	/webhooks.json	Register a webhook. Topics: orders/create, orders/updated, orders/paid, orders/fulfilled, orders/cancelled
GET	/webhooks.json	List registered webhooks for this app
DEL	/webhooks/{id}.json	Delete / deregister a webhook
POST	/fulfillments.json	Create a fulfillment record (mark as shipped) with tracking number and courier

🕒 RATE LIMITS — LEAKY BUCKET

Shopify uses a **leaky bucket** algorithm: bucket size 40 requests, refill rate 2 requests/second. Every response includes `X-Shopify-Shop-API-Call-Limit: used/max` (e.g. 12/40). On 429, wait for the `Retry-After` header value (seconds). For GraphQL, Shopify uses cost-based throttling (queries cost 1–1000 points).

Webhook Topics for Order Tracking

ORDERS/CREATE

Fires when a new order is placed — use for real-time order ingestion

ORDERS/UPDATED

Fires on any order change — status, shipping info, line items, notes

ORDERS/PAID

Fires when payment is captured — trigger COD confirmation or fulfilment workflow

ORDERS/CANCELLED

Fires on cancellation — trigger refund workflow or courier cancellation

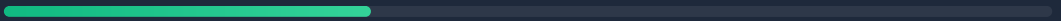
STRENGTHS

- ✓ Full customer PII always available (no masking)
- ✓ Real-time webhooks for orders, payments, fulfilments
- ✓ Line items included in single order response
- ✓ Simple token auth — no signing algorithm required
- ✓ Excellent documentation and SDK support
- ✓ REST + GraphQL both available
- ✓ 250 orders per request (vs Daraz's 100)

LIMITATIONS

- ✗ Access token shown only once — must be stored immediately
- ✗ Cursor-based pagination can be complex for deep paging
- ✗ Revenue data may be in non-PKR currencies
- ✗ Webhook secret management adds operational overhead

Complexity



Low



TCS Courier

Domestic Courier · Pakistan (established 1983) · 300+ cities covered

BEARER TOKEN

2-STEP AUTH

DEV + PROD URLs

DEV BASE URL

`https://devconnect.tcscourier.com`

PROD BASE URL

`https://ociconnect.tcscourier.com`

AUTH METHOD

Bearer Token
(cached, 2-step login)

PLATFORM ID (ZYRO DB)

3

ORDER LIST API

None — compute from Supabase

WEBHOOK SUPPORT

None

REQUIRED CREDENTIALS

`tcs_api_key`

→ Username / API Key provided by TCS after merchant registration

`tcs_api_password`

→ Password paired with the API Key for authentication

`tcs_cost_centre_code`

→ Cost Centre Code — identifies your account for shipment booking (required for all bookings)

Two-Step Authentication Flow

POST

`/auth/api/auth`
with apiKey +
password

Receive Bearer
Token

Cache token in
memory

Use

Authorization:
Bearer {token}
on all API calls

Re-authenticate
on 401

KEY API ENDPOINTS

POST

`/auth/api/auth`

Authenticate and receive Bearer token. Body: { userName, password }

POST

`/cod/api/v1/shipments`

Book a new shipment (CN creation). Requires costCentreCode, consignee details, weight, COD amount

GET

`/cod/api/v1/tracking/{cn}`

Track a shipment by Consignment Number (CN) — returns status events timeline

POST

`/cod/api/v1/shipments/cancel`

Cancel a booked shipment before pickup

GET

`/cod/api/v1/cities`

List all serviceable cities and their codes (needed for destination city field)

Shipment Booking Required Fields

CONSIGNEENAME

Customer full name

CONSIGNEEPHONE

Phone number (03xxxxxxxxx format)

CONSIGNEEADDRESS

Delivery address (full text)

COSTCENTRECODE

Your TCS merchant code

WEIGHT

Package weight in KG

CODAMOUNT

Cash-On-Delivery amount (PKR)

⚠ NO ORDER LIST API

TCS does not provide an endpoint to list all booked shipments. Order history and stats must be computed from your own Supabase database (filter by platform_id = 3). The TCS API is strictly for booking new shipments and tracking individual CNs.

STRENGTHS

- ✓ Pakistan's largest courier — 300+ city coverage
- ✓ Well-established merchant API with sandbox environment
- ✓ COD collection and remittance handled by TCS
- ✓ Real-time tracking events via CN number

LIMITATIONS

- ✗ No order list API — all history tracking must be done in Zyro DB
- ✗ Bearer token expiry requires caching strategy
- ✗ Separate Dev and Prod URLs — must switch on go-live
- ✗ Requires direct merchant registration with TCS

Complexity



Medium



Leopards Courier

Domestic Courier · Pakistan · 400+ cities covered · COD + Express

BODY CREDENTIALS

TEST MODE

CITIES API

BASE URL

`https://merchantapi.leopardscourier.com`

AUTH METHOD

API Key + Password in request body

TEST MODE

`enable_test_mode: "1"`

in body

PLATFORM ID (ZYRO DB)

4

ORDER LIST API

None — compute from Supabase

RESPONSE FORMAT

JSON with status code in body

REQUIRED CREDENTIALS

`leopards_api_key`

→ API Key provided by Leopards merchant portal

`leopards_api_password`

→ API Password paired with the key — both sent in every request body (no header auth)

KEY API ENDPOINTS

POST

`/api/v1/authenticate_user`

Verify credentials — call on initial connection to validate API key and password

POST

`/api/v1/add_packet`

Book a new shipment (creates packet). Returns packet tracking number

POST

`/api/v1/track_packet`

Track a packet by tracking number — returns full event history

POST

`/api/v1/cancel_packet`

Cancel a booked packet before pickup

POST

`/api/v1/get_cities`

List all serviceable cities with city codes — required for `destination_city` field in `add_packet`

POST

`/api/v1/get_packet_list`

Retrieve list of recent packets (limited to recent bookings, not full history)

Add Packet Request Body Structure

API_KEY

Your Leopards API key (sent in every request)

API_PASSWORD

Your API password (sent in every request)

ENABLE_TEST_MODE

"0" = live, "1" = test (no real booking)

CONSIGNEE_NAME

Customer full name

CONSIGNEE_PHONE

Phone number (03xxxxxxxxx)

DESTINATION_CITY

City code from `get_cities` endpoint

SHIPMENT_TYPE_ID

1=Parcel, 2=Document,
3=Cargo

PAYMENT_TYPE

1=Cash, 2=Credit

COD_AMOUNT

COD amount in PKR (0 if
prepaid)

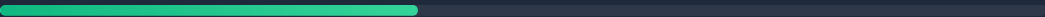
STRENGTHS

- ✓ 400+ cities — broader than TCS
- ✓ Test mode built into every API call — easy sandbox
- ✓ Cities API simplifies destination validation
- ✓ No separate auth step — credentials inline in body
- ✓ Real COD and express service options

LIMITATIONS

- ✗ Credentials in request body (not header) — less secure
- ✗ No order history / list API
- ✗ Response format uses body status codes (not HTTP status)
- ✗ No webhook support

Complexity



Low-Med



M&P Courier Services

Domestic Courier · Pakistan · Same-day & Next-day delivery · COD + Prepaid

REST API

API KEY AUTH

JSON

M&P (Methaq & Co / Movers & Packers) is one of Pakistan's growing courier networks with a focus on same-day and express delivery in major cities. Their merchant API allows e-commerce sellers to book, track, and manage shipments programmatically.

MERCHANT PORTAL

mnpcourier.pk /
portal.mnpcourier.pk

AUTH METHOD

API Key in request header or
body

RESPONSE FORMAT

JSON (success/error in body)

PLATFORM ID (ZYRO DB)

7 (suggested)

COVERAGE

Major cities + growing network

WEBHOOK SUPPORT

None currently

REQUIRED CREDENTIALS

mp_api_key

→ API Key issued by M&P merchant registration team

mp_account_number

→ M&P merchant account number (used in shipment booking)

KEY API ENDPOINTS

POST

/api/bookshipment

Book a new shipment. Returns tracking number (CN) for the parcel

GET

/api/trackshipment/{cn}

Track a shipment by CN number — returns current status and event history

POST

/api/cancelshipment

Cancel a booked shipment (must be before pickup)

GET

/api/getcities

Get list of serviceable cities — required for booking destination validation

POST

/api/getlabel

Generate shipping label PDF for a booked CN number

GET

/api/getshipments

List recent shipments for the account (limited date range)

Book Shipment Required Fields

CONSIGNEENAME

Customer name

CONSIGNEEPHONE

Contact number (03xxxxxxxxx)

CONSIGNEEADDRESS

Delivery address

CONSIGNEECITY

City name or code from getCities

CODAMOUNT

Cash on delivery amount (PKR)

WEIGHT

Package weight in KG

DESCRIPTION

Parcel/product description

ACCOUNTNUMBER

Your M&P merchant account number

i API ACCESS NOTE

M&P's merchant API is not publicly listed on their website. Sellers must contact M&P's merchant services team directly (via mnpcourier.pk or calling their helpline) to request API access credentials. The API is typically granted to merchants with a minimum monthly volume commitment. Documentation is provided upon onboarding.

STRENGTHS

- ✓ Same-day delivery in major cities (Karachi, Lahore, Islamabad)
- ✓ Simpler API structure than TCS
- ✓ Shipping label generation endpoint
- ✓ Growing city network with competitive rates
- ✓ Good for time-sensitive e-commerce deliveries

LIMITATIONS

- ✗ API access requires direct negotiation — not self-serve
- ✗ Limited public documentation available online
- ✗ Smaller city coverage vs TCS and Leopards
- ✗ No webhook or push notification support
- ✗ Monthly volume commitment may be required

Complexity



Medium



Pakistan Post

Government Postal Service · Nationwide · EMS + Registered Mail + Parcel Post

NO BOOKING API

TRACK ONLY

3RD-PARTY REQUIRED

Pakistan Post (Pak Post) is the government-operated postal service of Pakistan. It covers every corner of the country including remote areas not reachable by private couriers. However, **Pakistan Post does not offer a public merchant booking API**. Their digital infrastructure is limited to a basic tracking portal.

⚠ NO PUBLIC MERCHANT API

Pakistan Post has not released a public REST API for merchants. Shipment booking, label printing, and order management must be done manually through their counters or via Pakistan Post's dedicated corporate accounts. The only programmatic capability available is **tracking via scraping or third-party aggregators**.

OFFICIAL TRACKING PORTAL

track.pakpost.gov.pk

PROGRAMMATIC TRACKING

Via TrackingMore API (3rd party)

COURIER CODE (TRACKINGMORE)

pakistan-post

TRACKING FORMAT

13-char alphanumeric
(EE123456789PK)

BOOKING METHOD

Manual at post office counters

COVERAGE

All of Pakistan including remote areas

TrackingMore API Integration (Recommended Approach)

TrackingMore is a third-party parcel tracking aggregator that covers 1,200+ carriers including Pakistan Post. It provides a unified REST API to track shipments across all carriers.

TRACKINGMORE API CREDENTIALS

trackingmore_api_key

→ API Key from trackingmore.com developer portal (free tier available)

TRACKINGMORE ENDPOINTS FOR PAKISTAN POST

POST <https://api.trackingmore.com/v4/trackings/create> Register a tracking number. Body: { tracking_number, courier_code: "pakistan-post" }

GET https://api.trackingmore.com/v4/trackings/{courier_code}/{tracking_number} Get tracking events for a specific shipment

POST <https://api.trackingmore.com/v4/trackings/batch> Batch-track up to 40 tracking numbers in one request

TRACKINGMORE FREE TIER

100 trackings/month free, then per-tracking pricing

WEBHOOK SUPPORT

Yes — TrackingMore can push status updates via webhook

Pakistan Post Service Types

EMS (EXPRESS MAIL SERVICE)

Fastest — 1-2 business days. Trackable. For domestic and

REGISTERED MAIL / PARCEL

3-7 days. Trackable. Suitable for documents and light

MONEY ORDER

Cash remittance service — no API integration possible

international.

parcels.

STRENGTHS

- ✓ Widest coverage — every city, town, and village in Pakistan
- ✓ Most affordable shipping rates of all couriers
- ✓ Government-backed — reliable for remote areas
- ✓ International EMS coverage (190+ countries)

LIMITATIONS

- ✗ No public merchant booking API — manual booking only
- ✗ Tracking requires 3rd-party aggregator (TrackingMore)
- ✗ Slower delivery compared to private couriers
- ✗ No COD service (cash-on-delivery not supported)
- ✗ No real-time tracking updates — delays of hours or days

Integration

Tracking Only



DHL Express (MyDHL API)

International Express Courier · 220+ countries · OAuth 2.0

OAuth 2.0

Full API

International

Developer Portal

developer.dhl.com

API Name

MyDHL API

Base URL

https://express.api.dhl.com/mydhlapi

Auth Method

HTTP Basic Auth
(username +
password)

Sandbox URL

https://express.api.dhl.com/mydhlapi/test

Platform ID (Zyro DB)

8 (suggested)

Required Credentials

dhl_api_key	→ Username / API Key from MyDHL API dashboard (developer.dhl.com)
dhl_api_secret	→ Password / API Secret paired with the key — used in HTTP Basic Auth header
dhl_account_number	→ DHL account number (8-digit) — required for shipment booking and rate quotes

Authentication Method

DHL MyDHL API uses **HTTP Basic Authentication**. Every request includes the header: `Authorization: Basic base64(apiKey:apiSecret)`. No token exchange required — simpler than OAuth.

Key API Endpoints

POST	/shipments	Create a shipment. Returns AWB (Air Waybill) number, label PDF URL, and tracking number
GET	/tracking	Track shipments by AWB or shipmentTrackingNumber. Params: shipmentTrackingNumber
GET	/rates	Get rate quotes for a shipment. Params: origin, destination, weight, dimensions, account
POST	/pickups	Schedule a pickup from your location. Returns confirmation ID
DEL	/pickups/{dispatchConfirmationNumber}	Cancel a scheduled pickup
GET	/address-validate	Validate a shipping address and check DHL serviceability
GET	/reference-data/countries	List supported countries and their codes
GET	/reference-data/services	List available DHL Express service types (EXPRESS WORLDWIDE, EXPRESS 9:00, etc.)

Create Shipment — Key Request Fields

PLANNEDSHIPPINGDATEANDTIME

ISO 8601 datetime for planned pickup

PICKUP.ISREQUESTED

true = DHL picks up from your address

PRODUCTCODE

D (DHL Express Worldwide — most common for PK)

ACCOUNTS[].NUMBER

Your 8-digit DHL account number

CUSTOMERDETAILS.SHIPPERDETAILS

Sender address, name, postal code, country code

CUSTOMERDETAILS.RECEIVERDETAILS

Recipient address, name, country (international)

CONTENT.PACKAGES[].WEIGHT

Package weight in KG

CONTENT.DECLAREDVALUE

Customs declared value (for international shipments)

RATE LIMITS

DHL MyDHL API enforces rate limits per API key. Standard accounts: **500 requests/hour** per endpoint. Shipment creation is limited to burst protection — avoid more than 10 shipments/minute in bulk operations. DHL returns HTTP 429 with a Retry-After header when limits are hit.

✓ SANDBOX VS PRODUCTION

Append /test to the base URL for sandbox: <https://express.api.dhl.com/mydhlapi/test>. The same credentials work for both environments. Sandbox shipments do not create real AWBs and can be used for full integration testing including label generation and tracking simulation.

STRENGTHS

- ✓ Full API: booking, tracking, rates, pickup, label — everything in one
- ✓ Simple Basic Auth — no token management needed
- ✓ Sandbox environment mirrors production exactly
- ✓ 220+ country coverage for international shipments from Pakistan
- ✓ Real-time tracking with granular event data
- ✓ Rate quoting API — show customers live shipping cost
- ✓ Address validation to prevent failed deliveries

LIMITATIONS

- ✗ Premium pricing — significantly higher than domestic couriers
- ✗ International only — not cost-effective for domestic deliveries
- ✗ Requires DHL account number (corporate account signup)
- ✗ Complex shipment request payload (nested JSON objects)
- ✗ Customs declaration required for international parcels

Complexity



Medium

Comparison Matrix

All 7 integrations side-by-side across key technical dimensions

E-Commerce Platforms

Platform	Auth Method	Webhooks	Max/Request	Customer PII	Search	Complexity	Platform ID
Daraz	HMAC-SHA256 Signing	✗ None	100 orders	Masked (SDP req.)	Order ID only	High	5
Shopify	X-Shopify-Access-Token	✓ Full support	250 orders	Always visible	Name, email, order#	Low	6
WooCommerce	Basic Auth (Key+Secret)	✓ Push webhook	100 orders	Always visible	Name, email, phone	Low	1

Courier Services

Courier	Auth Method	Book API	Track API	List API	City API	COD	Coverage
TCS	Bearer Token (2-step)	✓ Yes	✓ Yes	✗ No	✓ Yes	✓ Yes	300+ cities
Leopards	Credentials in body	✓ Yes	✓ Yes	⚠ Limited	✓ Yes	✓ Yes	400+ cities
M&P	API Key in header/body	✓ Yes	✓ Yes	⚠ Limited	✓ Yes	✓ Yes	Major cities
Pakistan Post	None (manual only)	✗ No	⚠ Via TrackingMore	✗ No	✗ No	✗ No	All of Pakistan
DHL Express	HTTP Basic Auth	✓ Yes	✓ Yes	⚠ Limited	N/A	✗ No	220+ countries

Implementation Effort & Priority

Integration	Est. Dev Time	API Quality	Documentation	Business Impact	Recommended Priority
Shopify	2-3 days	Excellent	Excellent	Very High	P1 — NOW
TCS	3-4 days	Good	Moderate	Very High	P1 — NOW
Leopards	2-3 days	Good	Moderate	High	P2 — SOON
Daraz	4-6 days	Moderate	Moderate	Very High	P2 — AFTER SDP

INTEGRATION	EST. DEV TIME	API QUALITY	DOCUMENTATION	BUSINESS IMPACT	RECOMMENDED PRIORITY
DHL Express	3–4 days	Excellent	Excellent	Medium	P3 — LATER
M&P	2–3 days + negotiation	Moderate	Limited	Medium	P3 — LATER
Pakistan Post	1 day (tracking only)	Very Limited	None	Low	P4 — OPTIONAL

1 Shopify — Implement Webhooks First, Not Polling

Shopify's `orders/create` webhook delivers new orders in under 2 seconds. Do not build a polling loop — register the webhook on first connection and handle order ingestion purely via push. Store the webhook ID per user so it can be deregistered when the user disconnects. Use HMAC-SHA256 signature verification for every incoming webhook to prevent spoofed requests. This is the same pattern as WooCommerce and will bring the most value with the least polling overhead.

2 TCS + Leopards — Token Caching is Critical

TCS issues a Bearer token that expires. Cache this token in a server-side in-memory object keyed by user ID. On every API call, check if the cached token is still valid. On a 401 response, automatically re-authenticate and retry. For Leopards, credentials go in every request body — no caching needed, but this increases payload size and means secrets traverse the wire more often. Consider implementing AES-256 encryption on the stored credentials in Supabase as a security upgrade.

3 Daraz — Apply for Sensitive Data Privilege Immediately

The DataMoat system makes Daraz largely useless for CRM without the Sensitive Data Privilege (SDP). Apply for SDP on the Daraz App Console as soon as the integration is live — approval takes 2–4 weeks. Until SDP is granted, Daraz orders will show masked customer names and empty phone numbers. Design the UI to gracefully handle masked data (show "Data Protected — SDP Pending" rather than blank fields) so users understand the limitation.

4 Unified Phone Normalization (All Pakistani Couriers)

TCS, Leopards, and M&P all require phone numbers in `03xxxxxxxx` format (10 digits, no country code, no +92). Implement a single `normalizePhone()` utility that strips spaces, dashes, +92 prefix, and 0092 prefix, then pads to the correct format. Apply it to every customer phone before sending to any courier API. Failing to normalize causes silent booking failures — the courier API accepts the request but delivery attempts fail.

5 Pakistan Post — Use TrackingMore, Not Direct Integration

Pakistan Post has no merchant booking API and their tracking system is basic. Rather than investing in a custom scraping solution, integrate with **TrackingMore API** (trackingmore.com). A single TrackingMore API key lets you track Pakistan Post EMS, TCS, Leopards, DHL, and 1,200+ other carriers — useful as a fallback tracker for any courier. The free tier (100 trackings/month) covers initial usage; upgrade as volume grows. Store the TrackingMore key per-user or as a global system key depending on your pricing model.

6 DHL — Use for International Shipments Only

DHL Express rates are significantly higher than domestic couriers (PKR 1,500–3,000+ for a small parcel vs PKR 150–300 via TCS/Leopards). Position the DHL integration explicitly for international orders — filter by non-PK destination addresses and auto-suggest DHL when the customer address is outside Pakistan. The DHL rate quoting API (`/rates` endpoint) allows you to show live shipping cost to the user before booking, which is a strong UX differentiator.

7 Supabase Column Strategy — Per-User Credentials

All credentials must be stored per-user in the `users` table in Supabase — never in `.env.local` for multi-tenant use. The column naming convention already established: `{platform}_api_key`, `{platform}_api_password`, `{platform}_access_token`. Add columns incrementally via `SQL ALTER TABLE users ADD COLUMN IF NOT EXISTS` migrations — never require a full schema rebuild. For production security, consider Supabase Row Level Security (RLS) policies that prevent any user from reading another user's credentials.

8 Auto-Sync Architecture for Pull-Only Platforms

For Daraz (pull-only), implement a Vercel cron job at `/api/cron/daraz` that runs every 30 minutes. Query Supabase for all users with configured Daraz credentials, then pull and upsert orders per user. For Shopify and WooCommerce, the cron is only a fallback — rely on webhooks as the primary channel. Rate limit the cron to process 5 users per minute to avoid hitting Daraz's ~60 req/min threshold. Add a `last_synced_at` column per user to track when each user's data was last refreshed.

Recommended Supabase Columns Summary

COLUMN NAME	PLATFORM	DESCRIPTION
<code>daraz_app_key</code>	Daraz	App Key from Daraz Open Platform
<code>daraz_app_secret</code>	Daraz	App Secret (used for HMAC signing)
<code>daraz_access_token</code>	Daraz	Access Token from Self-Authorization
<code>shopify_store_domain</code>	Shopify	<code>mystore.myshopify.com</code>
<code>shopify_access_token</code>	Shopify	Admin API token (<code>shpat_XXXXX</code>)
<code>shopify_webhook_secret</code>	Shopify	Optional — for HMAC webhook verification
<code>tcs_api_key</code>	TCS	TCS username / API Key
<code>tcs_api_password</code>	TCS	TCS API password
<code>tcs_cost_centre_code</code>	TCS	Required for all TCS shipment bookings
<code>leopards_api_key</code>	Leopards	Leopards API key
<code>leopards_api_password</code>	Leopards	Leopards API password
<code>mp_api_key</code>	M&P	M&P API key (from merchant onboarding)
<code>mp_account_number</code>	M&P	M&P merchant account number
<code>dhl_api_key</code>	DHL	DHL MyDHL API username
<code>dhl_api_secret</code>	DHL	DHL MyDHL API password
<code>dhl_account_number</code>	DHL	DHL account number (8-digit)
<code>trackingmore_api_key</code>	Pakistan Post	TrackingMore API key for universal tracking

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